

DATA266 Lab 1 Report, Team 8

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DATA266 Lab 1, Team 8

Repository

<https://github.com/aneessaheba/DATA-266-Lab-1>

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1. Team ownership statement

Both members independently designed, coded, trained and evaluated their own models for all three tasks, so every number in this report exists twice, once per member, from two separate runs. Anees built the four block character level GPT for Task 1, the max pooling, LSTM and attention pooling classifiers for Task 2, and the six block CycleGAN for Task 3, and also set up the repository, the branching model, the data download script and the one command smoke test. Yashashree built the six block character level GPT for Task 1, the mean pooling, convolutional and bidirectional GRU classifiers for Task 2, and the nine block CycleGAN for Task 3. Both members submitted independently to the Kaggle competition. The comparison tables, the joint analyses and this report were written together from both sets of results.

2. Task 1, character level language model

2.1 Comparison table

	Anees	Yashashree
Transformer blocks	4	6
Embedding width	256	192
Attention heads	4	3
Head dimension	64	64
Feedforward width	1024	768
Context length	128	256
Dropout	0.10	0.15
Positional embedding	learned	learned
Normalisation	pre layer norm	pre layer norm
Vocabulary	91	101
Parameters	3,238,912	2,757,605
Optimiser	AdamW	AdamW
Learning rate	3e-4	3e-4
Weight decay	0.01	0.01
Warmup steps	1000	1000
Schedule	cosine decay	cosine decay
Batch size	64	64
Epochs	10	10
Seed	5330	4349
Validation cross entropy	0.7538	0.7077
Validation perplexity	2.1251	2.0293
Validation bits per character	1.0875	1.0210
Top 1 next character accuracy	0.7607	0.7759
Generalisation gap	-0.0303	-0.0009
Gradient norm mean	0.6679	0.6070
Gradient norm max	6.4868	5.4474
Loss spikes	0	0
NaN count	0	0
Distinct 3 at T = 0.8	0.7630	0.6663
Repeated 4gram at T = 0.8	0.1320	0.2310
Training tokens per second	37,528	610,618
Total training time	3,557 s	435 s
Hardware	Apple M4, MPS	RTX 4090

Neither member used a prebuilt transformer or attention module. Both wrote multi head causal self attention with explicit matrix multiplications and verified numerically that no position can attend to a future position.

2.2 Joint analysis

Strengths. Both models learned the TinyStories distribution to a similar degree, around 2.0 to 2.1 perplexity and roughly 76 to 78 percent next character accuracy, with no loss spikes and no non finite steps in either run. Both generalisation gaps are negative, meaning validation loss sits at or below training loss, so neither model overfitted in ten epochs.

Where they differ and what it shows. Yashashree's model reaches a better bits per character, 1.0210 against 1.0875, with 15 percent fewer parameters. The two candidate explanations are her greater depth and her longer context. Anees's failure analysis, written before the comparison, predicted that the 128 character context was the binding constraint because coherence broke down over spans longer than about two sentences. Her 256 character context reaching a better result on a smaller model is consistent with that, but depth and context changed together, so this is evidence rather than proof.

The result that surprised us. The model with the better validation loss produces the more repetitive text. At the same temperature of 0.8, Anees's repeated 4gram rate is 0.1320 against Yashashree's 0.2310, and his distinct 3 is 0.7630 against 0.6663. Perplexity measures how well a model predicts the next character given a true prefix. Repetition is a property of what happens when a model is fed its own output, which perplexity never observes. These are different questions and the ranking does not have to agree.

Limitations. The two vocabularies are not the same set. Anees found that the source text contained mojibake, UTF-8 that had been decoded as cp1252, and repaired it, which took his vocabulary from 101 characters to 91. Yashashree's run models the corrupted sequences as if they were real characters. Bits per character is still comparable because it is normalised per character either way, but a few of the characters in her run are artifacts. If anything this makes her result stronger, since her model carries that overhead and still predicts better.

Training times are not comparable. Apple M4 through MPS against an RTX 4090 is a hardware difference of roughly eight times, not a model difference.

What we would try next. One run of Anees's architecture at a 256 character context with everything else held fixed. That isolates context length from depth, which this comparison cannot do because both changed at once.

3. Task 2, Yelp Polarity sentiment classification

3.1 Comparison table

Test split, 38,000 reviews, balanced. Neither member used pretrained embeddings or a pretrained language model.

	Anees baseline	Anees exp A	Anees exp B	Yash baseline	Yash exp A	Yash exp B
Encoder	masked max pooling	2 layer LSTM	learned attention pooling	masked mean pooling	multi width CNN	bidirectional GRU
Vocabulary	25,000	25,000	25,000	30,000	30,000	30,000
Minimum frequency	3	3	3	2	2	2
Maximum length	320	320	320	256	256	256
Learning rate	2e-4	2e-4	2e-4	3e-4	3e-4	3e-4
Epochs	8	8	8	5	5	5
Seed	5330	5330	5330	4349	4349	4349
Parameters	2,412,545	4,297,217	3,224,962	3,856,770	5,181,890	6,057,026
Accuracy	0.8991	0.9502	0.9341	0.9348	0.9285	0.9469
Macro F1	0.8991	0.9502	0.9341	0.9348	0.9285	0.9469
ROC AUC	0.9643	0.9896	0.9819	0.9822	0.9818	0.9869
PR AUC	0.9657	0.9899	0.9822	0.9824	0.9823	0.9871
MCC	0.7981	0.9005	0.8683	0.8696	0.8579	0.8938
Brier	0.0756	0.0375	0.0490	0.0482	0.0533	0.0412
Expected calibration error	0.0392	0.0104	0.0024	see note	see note	see note
Training time	408 s	9,685 s	437 s	256 s	293 s	290 s
Peak memory	1.05 GB	1.26 GB	1.27 GB	0.16 GB	0.45 GB	1.80 GB
Hardware	Apple M4, MPS	Apple M4, MPS	Apple M4, MPS	RTX 4090	RTX 4090	RTX 4090

Note on calibration. Yashashree's reported expected calibration error of 0.4364, 0.4525 and 0.4629 cannot be correct alongside her Brier scores of 0.0482, 0.0533 and 0.0412. For a balanced test set the Brier decomposition gives Brier at least equal to reliability, and reliability is at least the square of the expected calibration error, so an error of 0.4629 would require a Brier of at least 0.214. All three of her models break that bound, so the calculation is wrong rather than the models being badly calibrated. Her Brier scores are sound and comparable to Anees's. The values are left in her `metrics_report.csv` for transparency but are not quoted here.

3.2 Significance tests

McNemar against each member's own baseline, on paired predictions over the same test reviews.

Model	Statistic	p value	Baseline right, model wrong	Model right, baseline wrong
Anees experiment_lstm	1087.52	1.1e-252	765	2,710
Anees experiment_attention_pool	599.20	4.1e-137	814	2,147
Yash experiment_bigru	109.88	1.0e-25	725	1,184
Yash experiment_cnn	22.74	1.9e-06	1,376	1,136

The convolutional model is the one to read carefully. Its result is statistically significant but the counts run against it: the baseline is right and the CNN wrong on 1,376 reviews, while the reverse happens on only 1,136. It is significantly **worse** than its own baseline.

3.3 Joint analysis

Strengths. Both members' best models are recurrent, and both beat their own baselines by margins that are not close to chance. The best model across the team is Anees's LSTM at 95.02 percent, with Yashashree's bidirectional GRU at 94.69 percent and 29 percent more parameters. That both of us independently arrived at a recurrent model as the strongest is the clearest shared result: on this dataset, reading word order beats every order free method either of us tried.

The most useful disagreement is between the two baselines. Yashashree's mean pooling reaches 93.48 percent where Anees's max pooling reaches 89.91, a gap of 3.6 points between two models that differ only in how they collapse positions into one vector. Max pooling takes the single most extreme activation per dimension, so one strong word can decide a review. Mean pooling accumulates evidence across every token. For sentiment that matters, because a long review usually carries many mild signals rather than one decisive word. Anees's own error review supports this from the other direction: eight of his twenty errors turn on a contrast or a concession, which is exactly where one extreme token overrides the balance of the text. Had we known this before running, mean pooling would have been the better baseline for both of us.

Weaknesses. Yashashree's convolutional model is worse than her own baseline despite 34 percent more parameters. A convolution only reaches as far as its widest kernel, so a negation separated from its target by several words is still out of range, while mean pooling at least accumulates every token evenly. Anees's attention pooling was still improving at its final epoch, validation loss falling 0.1849 then 0.1817 then 0.1769, so its 93.41 percent is a lower bound rather than a ceiling. His LSTM began overfitting from epoch 7, with training loss falling while validation loss rose, and was saved by selecting on validation loss.

A methodological finding worth recording. Yashashree's error review initially concluded that negation was her main failure mode, because 16 of her 20 sampled errors contained negation. Her own slice metrics refute it: reviews containing negation fail at 0.0532 against 0.0531 for the test set as a whole, so there is no effect at all. The explanation is the base rate, since 61.2 percent of all test reviews contain negation, and a binomial test on 16 of 20 against that base rate gives $p = 0.063$. A hand inspection of 20 errors is a good way to generate a hypothesis and a bad way to test one, because an error sample carries no information about the base rate it was drawn from. The corrected analysis is in her `failure_analysis.md`.

Limitations. Training times are not comparable across Apple MPS and an RTX 4090. Anees ran 8 epochs and Yashashree 5, so the comparison between our runs is not purely architectural. Anees's

LSTM is not bit reproducible on MPS: across two runs its best epoch moved from 5 to 6, while his other two models reproduced exactly, so the cause is the LSTM kernels rather than the seeding.

What we would try next. Mean pooling as the baseline for both members, and a length ablation at 512 tokens, which Yashashree's slice metrics motivate: long reviews are her worst slice at 0.0565 against 0.0492 for medium, and all five of her sampled long review failures sit at exactly her 256 token limit.

4. Task 3, CycleGAN style transfer

4.1 Comparison table

	Anees	Yashashree
Generator	resnet, 6 residual blocks	resnet, 9 residual blocks
Base filters	80	64
Discriminator	PatchGAN, 16 by 16	PatchGAN, 16 by 16
Normalisation	instance	instance
Adversarial loss	least squares	least squares
Parameters per generator	12,239,363	11,378,179
Total parameters	30,008,200	28,285,832
Image size	256	256
Batch size	1	1
Epochs	50	50
Learning rate	2e-4	2e-4
Lambda cycle	10.0	10.0
Lambda identity	2.5	5.0
Schedule	constant to 25, then linear decay	constant to 25, then linear decay
Seed	5330	4349
FID, Monet to photo	178.94	189.22
FID, photo to Monet	86.59	92.39
KID, Monet to photo	0.0328	0.0431
KID, photo to Monet	0.0075	0.0122
Density, Monet to photo	0.9333	0.6222
Density, photo to Monet	0.5420	0.4163
Coverage, Monet to photo	0.0117	0.0070
Coverage, photo to Monet	0.7267	0.6867
Cycle L1, Monet side	0.1647	0.1411
Cycle L1, photo side	0.0949	0.0962
Content cosine, Monet to photo	0.7348	0.8011
Content cosine, photo to Monet	0.8658	0.7463
LPIPS, Monet to photo	0.3556	0.3620
LPIPS, photo to Monet	0.3820	0.4177
Non finite steps	0	0
Training time	32,198 s, 8.94 h	18,051 s, 5.01 h
Throughput	19.76 img/s	35.26 img/s
Peak memory	1.568 GB	1.563 GB
Hardware	RTX 4090	RTX 5090

	Anees	Yashashree
Kaggle public score	-52.8390	-54.6990

The competition takes a one row `submission.csv` with ID, FID and MiFID, scored by the instructor's evaluation script, and the leaderboard shows the negative mean of the two values. A less negative score is better. Anees submitted FID 105.2606 and MiFID 0.4175; Yashashree submitted FID 108.9811 and MiFID 0.4171.

The FID values in the table differ from the submitted ones because they are not the same measurement. Each member's notebook compares 300 real Monets against all generated images using its own Inception preprocessing, while the instructor's scorer subsamples both sides to 300, sorts by filename and resizes differently. Only the scorer's version is comparable across the class.

4.2 Joint analysis

Strengths. Both runs completed 50 epochs with zero non finite steps, no loss spikes and no sign of either discriminator overpowering its generator. Discriminator losses settled near 0.17 and 0.16 respectively and stayed there. Both models translate photographs into recognisably Monet like images and both reconstruct the round trip closely enough to confirm that cycle consistency is holding.

The comparison separates cleanly along one hyperparameter. The two runs used the same data, epochs, resolution, learning rate and schedule. Three things differ: the seed, the generator shape, and the identity weight. Anees is better on every FID and KID number in both directions. Yashashree is better on cycle reconstruction for the Monet direction, 0.1411 against 0.1647, and on content similarity, 0.8011 against 0.7348. That is exactly the trade the identity term controls. Identity loss penalises changing an image that is already in the target domain, which mostly means holding the colour palette fixed. At 2.5 the generator is freer to repaint, which the distribution metrics reward. At 5.0 the palette is held, which keeps the round trip faithful and leaves outputs looking closer to the source photograph.

This is visible as well as measurable. The same input photograph of a lighthouse is repainted by Anees's model and comes back from Yashashree's model still reading as a photograph with brush texture applied over it.

Weaknesses. Both models paint fake artist signatures into the lower part of the frame. Real Monets are signed, so a corner mark is evidence of authenticity to the discriminator and the generator learns to produce one. Measured over each member's 30 validation translations by comparing gradient energy in the bottom strip against the rest of the image, the artifact appears in 7 of 30 of Anees's outputs and 16 of 30 of Yashashree's. No automatic metric here penalises it, and both members' Kaggle submissions contain it.

Both runs also show grid structured artifacts in flat regions such as open sky. Both generators upsample with transposed convolutions using a kernel size of 3 and a stride of 2, and because 3 does not divide evenly by 2 some output pixels receive more contributions than others at a fixed spacing.

Limitations. Three variables changed at once between the two runs, so neither member can attribute the difference to the identity weight alone. The Monet to photo direction is weak for both, FID 178.94 and 189.22, partly because inventing photographic detail is harder than applying a style, and partly because it is measured on only 30 generated images against 7,038 real photographs, which depresses coverage to 0.0117 and 0.0070 and makes that figure less reliable than the reverse direction. Both runs used 50 epochs where the CycleGAN paper uses 200 for this task, and both members' cycle losses were still falling slowly at the end. Training times are not comparable across an RTX 4090 and an RTX 5090.

What we would try next. A single run with only the identity weight changed, holding the generator shape and the seed fixed, which would turn the correlation we observed into an attribution. Replacing the transposed convolutions with nearest neighbour upsampling followed by a plain convolution, which is the standard fix for the grid artifact. And masking signatures out of the training paintings, to test whether the signature artifact is learned from them as we believe.

5. Failure and error analysis

Full write ups with every snippet are in each member's `failure_analysis.md`. The cases below are quoted from those files.

5.1 Task 1, Anees

Repetition, greedy decoding. “One day, Tom went to the park with his mom. He saw a big storm on the ground. He wanted to see what was inside. He wanted to see what was inside.” The same sentence appears twice verbatim. This sample has the worst repetition of the twelve generated, 0.316 against 0.104 for the most diverse. Greedy decoding is deterministic, so once the model returns to a similar hidden state the same continuation becomes most probable again and nothing can break the loop.

Loss of coherence. “They saw a big boy who was very sad and angry. They wanted to show her mommy how to be careful when they saw the boy was so happy to have a new friend.” The boy is sad and angry then happy within one sentence. The grammar is correct throughout; what fails is state tracking, because the model predicts from the previous 128 characters only and has no representation of facts already asserted.

Hallucination at T = 1.2. “She loved playing outside and majwming running around ... Carefully, here no hourse. She saw her mom radious.” Three strings are not English words. This is specific to character level modelling: a word level model can only emit words from its vocabulary, while a character level model can reach any letter sequence.

5.2 Task 1, Yashashree

Broken grammar and semantic inconsistency. “She had a big smile on her house and a special wind.” A smile belongs to a person, not a house, and the wind is not connected to the sentence. The model learned common words and patterns without learning relationships between objects and actions.

Loss of coherence and abrupt topic change. “Every day, she saw a big tree with lots of money. The pink was a big pretty girl with a playground beautiful pla” The sentence introduces money, a pink girl and a playground with no transition, and ends incomplete.

Repetition and semantic inconsistency. “They saw a big tree with a big rock. The tree was filled with a tall green ball on the ground.” The model repeats “big”, “tree” and “ground” and describes a tree filled with a ball.

Her closing observation is the one worth carrying forward: the model reaches 77.6 percent next character accuracy while still producing these failures, because next character accuracy can be high even when the errors that occur damage the meaning of a longer story.

5.3 Task 2, both members

Both members reviewed 20 of their own errors in the four categories the brief requires: five confident false positives, five confident false negatives, five near threshold cases and five from a slice.

Anees, twenty errors. Eight turn on a reversal or a concession, four on mixed or hedged sentiment, two on third party narrative, two on vocabulary gaps, two on truncation and two look like wrong gold labels. His largest group makes contrast marker weighting the first fix to try. His slice numbers contradicted the intuitive hypothesis: the truncated slice has a **lower** error rate than the long slice for all three of his models, so raising the length limit is the obvious fix and the wrong one.

Yashashree, twenty errors. Five involve negation over a long span, five mixed or aspect split sentiment, four truncation at 256 tokens, three sarcasm, two temporal reversal and one too little signal. As recorded in section 3.3, her negation hypothesis did not survive her own slice metrics, while truncation did: all five of her long review failures sit at exactly her 256 token limit and long reviews are her worst slice.

Taken together, the two reviews disagree about truncation and the disagreement is explained by the limits themselves. Anees truncates at 320 tokens and sees no truncation effect; Yashashree truncates at 256 and sees one. That is consistent with the conclusion rather than against it: the cost of truncation appears somewhere between 256 and 320 tokens on this dataset.

5.4 Task 3, both members

Both members documented three visual cases with the images attached in their folders.

Anees. A hallucinated artist signature in `validation_B2A/0000.png`, measured in 7 of 30 samples. Repeated blobs and colour banding across a flat sky in `validation_B2A/0007.png`, from the transposed convolution upsampling. A colour shift in cycle reconstruction, where Monet's haystacks return with the composition intact but the palette moved from warm pink to cool blue.

Yashashree. Under stylisation in `validation_B2A/0000.png`, where the output keeps the photograph's structure, lighting and colour with brush texture laid over it. A bright blue and red checkerboard block in `validation_B2A/0007.png`. And her cycle reconstruction of a valley with poplars and cottages, included as the counterexample because it returns almost exactly, which is the same identity weight that caused her first case.

6. Evidence trail

Every number in this report is traceable to a committed file.

Evidence	Location
Raw training logs, unedited	reproducibility/raw_logs/
Manifests, six, one per member per task	reproducibility/manifests/
Task 1 metrics	task1_llm/<member>/metrics_report.csv
Task 1 loss curves	task1_llm/anees_saheba/outputs/loss_curves.png , task1_llm/yashashree_shinde/outputs/task1_loss_curves.svg
Task 1 generated samples	task1_llm/<member>/outputs/
Task 1 checkpoints	task1_llm/anees_saheba/checkpoints/best_model.pt , task1_llm/yashashree_shinde/checkpoints/task1_best_model.pt
Task 2 metrics	task2_sentiment/<member>/metrics_report.csv
Task 2 slice and significance	task2_sentiment/anees_saheba/metrics_report.csv , task2_sentiment/yashashree_shinde/task2_slice_metrics.json and task2_mcnemar_results.json
Task 2 error reviews	task2_sentiment/<member>/ error review csv and failure_analysis.md
Task 2 checkpoints	task2_sentiment/<member>/checkpoints/ , three per member
Task 3 metrics	task3_gan/<member>/metrics_report.csv and outputs/metrics/full_metrics_report.json
Task 3 training curves	task3_gan/<member>/outputs/figures/training_curves.png
Task 3 sample translations	task3_gan/<member>/outputs/validation_A2B , validation_B2A , validation_cycle_A , validation_cycle_B
Task 3 checkpoints	task3_gan/<member>/checkpoints/generators_epoch_050.pt
Kaggle submissions	task3_gan/<member>/outputs/submission.csv
Reproduction script	scripts/smoke_test.sh , one command, runs Task 1 end to end

Checkpoints are tracked with Git LFS. Optimiser state was removed from Yashashree's Task 1 and Task 2 checkpoints before committing, which reduced them from 206 MB to 70 MB without affecting the weights, since Adam moment buffers are needed only to resume training.

7. Known gaps

Stated rather than hidden.

The human audit is not complete for either member. Both `human_audit_30_samples.csv` files are 30 row templates with empty rater columns. The notebooks deliberately do not generate ratings. This is the only part of the Task 3 evaluation that would catch the signature artifact, since no automatic metric penalises it.

Yashashree's Task 1 and Task 2 runs captured no raw training logs. Her notebooks did not write them and they cannot be reconstructed after the fact. Her Task 3 log is present. Both members' manifests record this.

Yashashree's Task 2 outputs folder is empty, so there are no loss curves or confusion matrices for her three classifiers.

Yashashree's Task 2 calibration numbers are wrong, as set out in section 3.1, and need recomputing.

Anees's Task 3 checkpoint carries a stale config field recording nine residual blocks where the weights are unmistakably six at 80 filters, 12,239,363 parameters per generator. The YAML config and the write ups match the actual weights.

8. References

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L. and Polosukhin, I. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems* 30.

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Zhu, J.-Y., Park, T., Isola, P. and Efros, A. A. (2017). Unpaired Image to Image Translation using Cycle Consistent Adversarial Networks. *IEEE International Conference on Computer Vision (ICCV)*.

Naeem, M. F., Oh, S. J., Uh, Y., Choi, Y. and Yoo, J. (2020). Reliable Fidelity and Diversity Metrics for Generative Models. *International Conference on Machine Learning (ICML)*. Source of the density and coverage metrics used in Task 3.